

1. The "Clean Slate" Restart

If Vivado is still giving/showing weird 'X' or 'Z' values, do this first:

- **Close the Simulation Tab:** Click the '**X**' on the "Untitled 1" or "Waveform" tab.
- **Reset the Simulation:** In the **Flow Navigator** on the left, right-click **Run Simulation** and select **Reset Behavioral Simulation**. This clears the memory without restarting the whole app.

2. Manual Input Setup (The "Force" Values)

Once the waveform window opens again, your signals will likely be blank or 'Z'. Follow this exact order to set the initial state:

- **Step A: The Clock (Pulse)**
 - Right-click clk in the **Objects** window > **Force Clock**.
 - **Leading Edge:** 0
 - **Trailing Edge:** 1
 - **Period:** 10ns
 - *Why:* This gives the FSM a "heartbeat" so it can actually move between states.
- **Step B: The Reset (Parking the System)**
 - Right-click reset > **Force Constant**.
 - **Value:** 1
 - *Why:* This forces the FSM into the IDLE (00) state immediately.
- **Step C: The Sensors (Default Off)**
 - Right-click payment, exit_sensor, and obstruction one by one > **Force Constant**.
 - **Value:** 0 for all of them.

3. Running the Initial State

1. Click the **Run For** button (the icon with a "T" and a play button) or press **F5**.
2. Look at gate_state. It should now be a solid **0** (IDLE).
3. **Release the Reset:** Right-click reset > **Force Constant** > Value: **0**.
4. Click **Run For** again.

The system is now "Armed" and waiting for your inputs.

4. Testing the "Jump"

To see the gate move like you did before:

1. Right-click payment > **Force Constant** > Value: **1**.
2. Click **Run For**.
3. Observe the gate_state jump from **0** to **1** (OPENING) or **2** (OPEN).

How to Manually Simulate the "Safety Check"

1. **Trigger the Gate:**
 - Set payment to 1.
 - Click **Run** (F5). The gate_state should move to 01 (OPENING) or 10 (OPEN).
2. **Simulate the Obstruction:**
 - Right-click obstruction > **Force Constant** > Value: 1.
 - Click **Run** (F5).
3. **Verify the Damage Prevention:**
 - Check your gate_state. In a well-coded FSM, the gate should immediately stop, reverse back to OPENING, or stay OPEN regardless of any other sensors.
 - **Pro Tip:** While obstruction is 1, try forcing exit_sensor to 1. If your code is safe, the gate **should not move** to CLOSING or IDLE while the obstruction is detected.

Why didn't we use those Github links?

As we discussed, those repositories (UART, Motor Control, Traffic Light) are for much more advanced stages.

- **UART** is for sending text to a computer screen.
- **Motor Control** is for spinning a physical motor.
- **Traffic Light** is just a different version of what we already built.